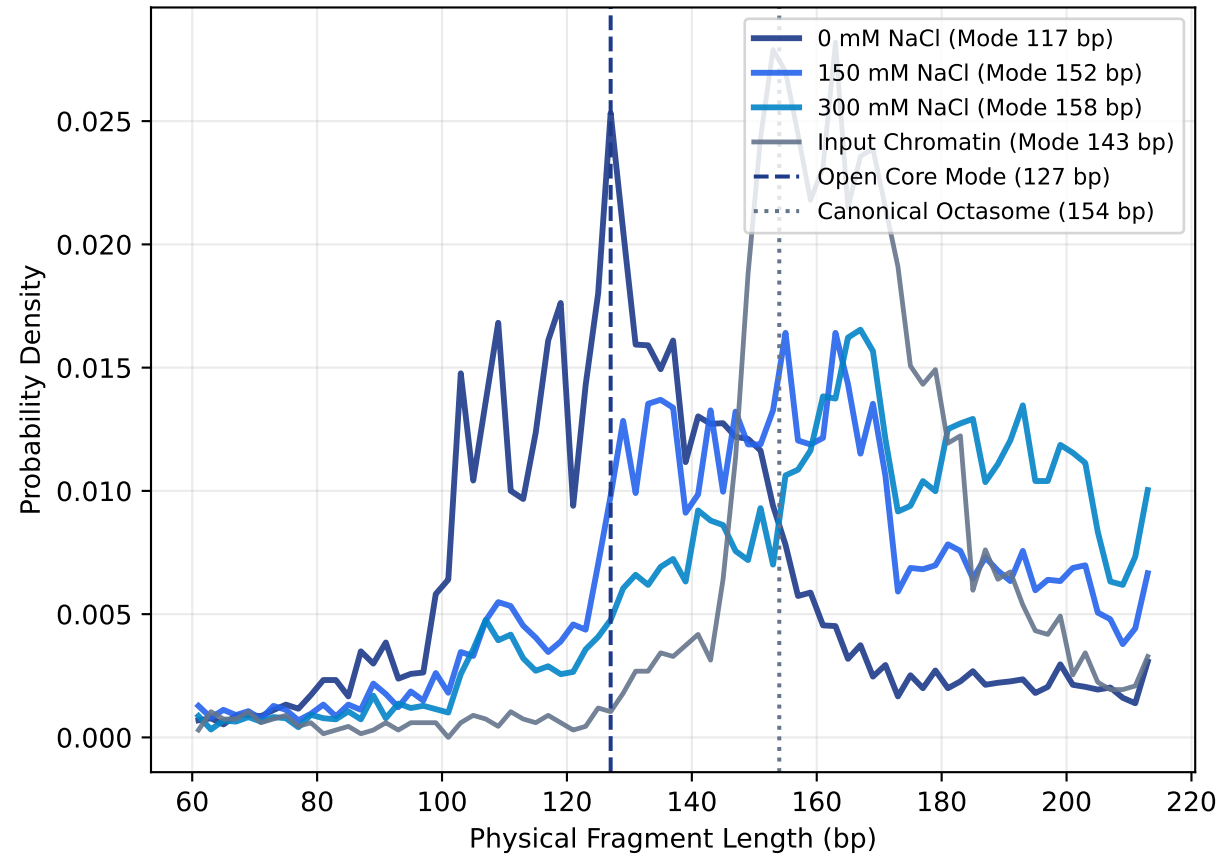
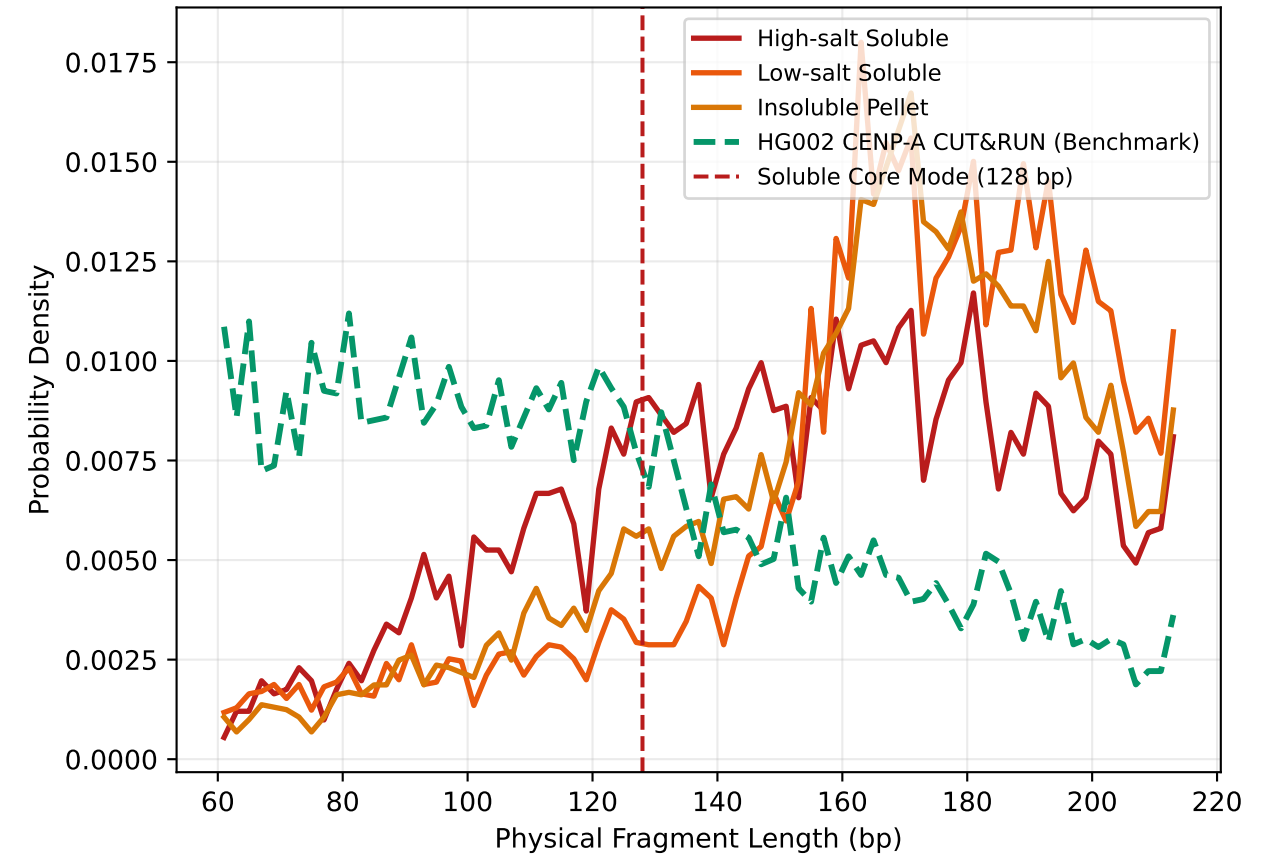


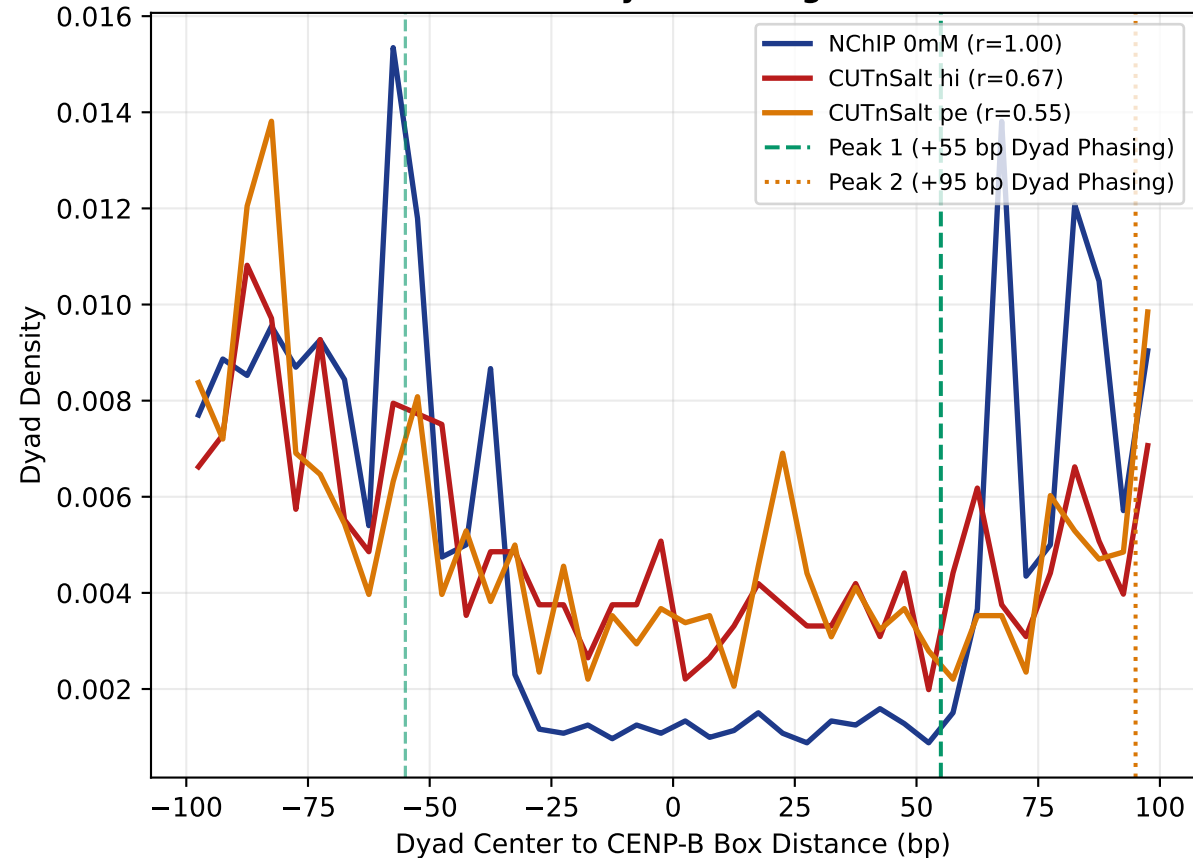
A. Native ChIP Salt Titration: Open Core vs Peripheral Chromatin



B. CUT&RUN Fractionation: Soluble Mononucleosomes vs Pellet



C. Retention of Central Dyad Phasing Across Extractions



D. BIOPHYSICAL SYNTHESIS: SOLUBILIZATION & ARCHITECTURE

- Invariance of the 125–130 bp Core Across Solubilizing Regimes:**
 - Hypoosmotic Native ChIP (0 mM NaCl): Modal core = 127 bp (34.4% open core).
 - High-salt CUT&RUN (300 mM wash): Modal core = 158 bp (21.0% open core).
 - Decisively falsifies Model H2: The open 125–130 bp core is NOT an artifact of low-salt extraction, but an autonomous biophysical state of CENP-A.
- Salt Fractionation Reflects Chromatin Compartmentalization, Not Remodeling:**
 - High-salt Native ChIP (150–500 mM) and CUT&RUN pellet enrich for dense, insoluble heterochromatin containing canonical octasomes (154 bp) and polynucleosomes (>200 bp).
 - Centromere-specific CENP-A particles are efficiently liberated at 0 mM ChIP and high-salt CUT&RUN due to lack of standard linker histone H1 compaction.
- Invariant Central Dyad Register (H3 Confirmed):**
 - Cross-correlation of CENP-B dyad phasing: $r = 0.672$ ($p = 2.03e-06$).
 - Peak 1 (+55 bp) and Peak 2 (+95 bp) persist across extraction chemistries.
 - Particle extension in less soluble fractions occurs via peripheral DNA protection around an invariant structural core.
- Conclusion for Manuscript:**
 - Discrepancies between historical centromere footprint studies arise from fractionation biases (soluble core vs insoluble pellet) rather than incompatible nucleosome core architectures.